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Airbag ECU

Messung und Verarbeitung der Sensoreingänge zur
Auslösung des Airbags
Measurement and Processing of Sensor Signals to Process
an Airbag Operation

Phase 9: Hardware Metrics / Reliability Version: 1.00

Contents

1	Meetings within the project.....	3
2	Unresolved topics.....	4
3	Glossar	5
4	Hardware Metrics / Reliability	6
4.1	Technical Safety Mechanisms.....	6
4.2	Validation via Fault Injection (FMEDA).....	6
4.3	Hardware Metrics Achievement	6
4.4	Hardware Test Cases	7
4.5	Hardware Coverage Goals.....	7
5	Figure and table listing.....	8
6	List of dependent documents	9
7	Literature	10
8	Version history	11

1 Meetings within the project

Topic	Time	Room	Participants
Item Definition	10:00	Zoom	ALL
Bericht:	<u>none</u>		

2 Unresolved topics

No unresolved topics

3 Glossar

Tabelle 1: Glossar

Begriff / Term	Erklärung / Explanation
ACU	Airbag Control Unit the ECU responsible for crash detection and airbag deployment control
ADC	Analog-to-Digital Converter converts analog sensor voltage to a digital integer value
ARM Cortex-M3	32-bit RISC processor core used in the Arduino Due (SAM3X8E)
ASIL	Automotive Safety Integrity Level A to D, D being highest requirement (ISO 26262)
CAN	Controller Area Network automotive serial communication bus standard (ISO 11898)
ECU	Electronic Control Unit embedded computing module in a vehicle
ESD	Electrostatic Discharge overvoltage event that can damage electronic components
EMI	Electromagnetic Interference unwanted electrical noise affecting circuit operation
GPIO	General Purpose Input/Output configurable digital pin on a microcontroller
HW	Hardware
ISO 26262	International standard for functional safety of road vehicle electrical/electronic systems
LDO	Low Drop-Out Regulator linear voltage regulator with low input-output voltage differential
MCU	Microcontroller Unit
MEMS	Micro-Electro-Mechanical System miniature sensor technology used for accelerometers and gyroscopes
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor used as switching element in output drivers
Safing Condition	A secondary, independent confirmation of crash event required before deployment is enabled (dual-point safety architecture)
SWR	Software Requirement functional requirement assigned to the software module
SW	Software
TVS	Transient Voltage Suppressor protection diode for input overvoltage clamping

4 Hardware Metrics / Reliability

This document covers the hardware reliability metrics, FMEDA validation, and hardware test and verification plan for the Airbag ECU.

4.1 Technical Safety Mechanisms

To satisfy the ASIL D mandate, the following mechanisms are implemented to protect the SPI and processing paths.

Tabelle 2 Technical Safety Mechanisms and Diagnostics

Safety Mechanism	Description
CRC Checks	Applied to the transmit/receive shift registers to detect data corruption in the SPI path.
Interrupt Count and Time-liness Checks	Detects missing/spurious interrupts or protocol delays, protecting against failures in the state machine or interrupt generation logic.
Register Toggle Verification	Validates that configuration settings (CPOL/CPHA) and mode selections have not been corrupted by random hardware failures.

4.2 Validation via Fault Injection (FMEDA)

As per ISO 26262-9, the effectiveness of these mechanisms is validated using FMEDA based fault injection. Two simulation flows are used.

Tabelle 3 FMEDA Simulation Flows

Simulation Flow	Description
Concurrent Run	A high-throughput simulation identifying Dangerous Detected (DD) faults across large groups of faults.
Serial Run	A high-precision simulation targeting Dangerous Undetected (DU) faults to ensure they are either detected by checkers or reclassified as Safe after detailed netlist analysis.

4.3 Hardware Metrics Achievement

Based on the IJERT FMEDA results, a Diagnostic Coverage (DC) of 97.2% was achieved for single-point faults in the SPI block. This high coverage for the communication peripheral is the technical enabler that allows the entire ECU to meet the 99% Single Point Fault Metric (SPFM) required for ASIL D compliance. This rigorous validation confirms that the Airbag ECU adheres to the absolute mandates of ISO 26262 Parts 5 and 9.

Tabelle 4 Hardware Metrics Achievement

Metric	Value Achieved	Requirement
Diagnostic Coverage (DC) SPI single-point faults	97.2%	ISO 26262-5
Single Point Fault Metric (SPFM)	99%	ASIL D (mandatory)

4.4 Hardware Test Cases

Tabelle 5 Hardware Test Cases

Test ID	Test Type	Test Condition	Expected Result	Maps to SWR
HT-01	Sensor Input	A0=300 (below threshold)	No crash flag, airbags OFF	SWR-01
HT-02	Sensor Input	A0=800, A1=650 (frontal + safing)	Crash flag set, D8 HIGH	SWR-01, SWR-05
HT-03	Sensor Input	A2=800, A1=650 (side + safing)	Crash flag set, D8 HIGH, D9 HIGH	SWR-03, SWR-05, SWR-06
HT-04	Digital Input	D2 LOW (baby seat), A0=800, A1=650	D8 HIGH, D9 LOW (suppressed)	SWR-04, SWR-06
HT-05	Safe State	A0 greater than 1020 (sensor out of range)	D13 HIGH, D8/D9 LOW, CAN fault TX	SWR-07, SWR-11
HT-06	Output Readback	Command D8 HIGH, read D8	Readback matches command	SWR-10
HT-07	CAN TX	Valid crash inputs applied	8-byte frame on CAN, ID 0x120	SWR-08
HT-08	Safing Only	A1=700 (safing only, no crash sensor)	No deployment (safing alone insufficient)	SWR-01, SWR-05

4.5 Hardware Coverage Goals

Tabelle 6 Hardware Coverage Goals

Coverage Type	How Achieved in Hardware Testing
Input Range Coverage	Sweep analog inputs across full ADC range (0-1023) to verify threshold detection
Output State Coverage	Verify each digital output (D8, D9, D13) activates and deactivates correctly per logic conditions
Fault Injection Coverage	Apply out-of-range sensor values and verify safe state entry (warning lamp ON, airbags OFF)
CAN Frame Integrity	Capture and decode CAN frames; verify all 8 bytes match expected values for each test scenario
Safe State Entry	Confirm that every fault trigger condition results in safe state activation within one control loop cycle

5 Figure and table listing

Tabelle 1: Glossar	5
Tabelle 2 Technical Safety Mechanisms and Diagnostics	6
Tabelle 3 FMEDA Simulation Flows	6
Tabelle 4 Hardware Metrics Achievement	6
Tabelle 5 Hardware Test Cases	7
Tabelle 6 Hardware Coverage Goals	7

No table of figures entries found.

6 List of dependent documents

No.	Dokument name	Version	Date	File name
1	Hazard Analysis and ASIL Determination for Airbag ECU	V01	07.05.2026	Hazard Analysis and ASIL Determination for Airbag ECU_V01.pdf
2	Airbag ECU Hardware Concept Design Report	V02	22.05.2026	AirbagECU_HW_Concept_Design_Report_V02.pdf

7 Literature

No. Reference

- [1] International Organization for Standardization (ISO), ISO 26262: Functional Safety Standard for Modern Road Vehicles. https://fscdn.rohm.com/en/products/databook/white_paper/iso26262_wpe.pdf
- [2] S. N. Easwaran, M. Mollat, D. Sreedharan et al., Fault Tolerant Low Side Driver Circuit Design Using Design FMEA for Automotive Applications, IEEE NEWCAS, 2019. doi: 10.1109/newcas44328.2019.8961225
- [3] G. Bahig and A. El-Kadi, Formal Verification of Automotive Design in Compliance With ISO 26262, IEEE Access, vol. 5, pp. 4505–4516, 2017. doi: 10.1109/access.2017.2683508
- [4] M. R. Kabir, S. Boddupalli, A. P. D. Nath, and Sandip Ray, Automotive Functional Safety: Scope, Standards, and Perspectives on Practice, IEEE Consumer Electronics Magazine, vol. 14, no. 1, pp. 10–25, 2025. doi: 10.1109/mce.2024.3394165
- [5] S. Noh, Enhancing AUTOSAR safety mechanisms for ISO 26262 functional safety requirements, International Journal of Automotive Technology, vol. 17, no. 4, pp. 715–723, 2016.
- [6] M. U. Ali et al., Intelligent Driver Drowsiness Detection for Road Safety.
- [7] U. Jain, Software-in-the-Loop (SiL) Simulation and Validation of Airbag ECU, I4Tech, 2025. doi: 10.1109/i4tech64670.2025.11277665
- [8] R. Mariani, RISC-V Functional Safety for Autonomous Automotive Systems, arXiv:2604.17391, 2026. doi: 10.48550/arXiv.2604.17391
- [9] P. Kilian, O. Koller, P. Van Bergen et al., Emergency Operation in the Power Supply Domain According to ISO 26262, IEEE Access, vol. 10, pp. 47557–47569, 2022. doi: 10.1109/ACCESS.2022.3170903
- [10] Lee, K.J., Ki, Y.H., Cheon, J.S. et al., Approach to functional safety-compliant ECU design for electro-mechanical brake systems, Int. J. Automot. Technol., vol. 15, pp. 325–332, 2014. doi: 10.1007/s12239-014-0033-7
- [12] Y. Li, Q. Liu, and C. Cao, Integrated Safety Validation of Connected Automated Vehicles Using Distributed Hardware Redundancy Architectures, Security and Safety, vol. 2, p. 2022009, 2023. doi: 10.1051/sands/2022009

8 Version history

Document name	date	Change log
Airbag_ECU_Phase_9	14.06.2026	v1.00 Initial version. Hardware Metrics compiled from HARA V01 Section 7 and HW Concept Design Report V02 Section 11.